

ground-score: an AI support agent for @AmazonHelp, and how much to trust it

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github.com/kavinnandha/ground-score · full decision log in DECISIONS.md

On the numbers. Every figure here is produced into `results/` by the code in the repo. Section 3.1 is complete: it needs no hand labels, so it has run. The baseline table (3.2) and the full failure ranking (4) are pending because the golden set is sampled but not yet adjudicated. The framing, method and limitations sections do not depend on that run.

1. Problem framing

What "good" means for this brand

@AmazonHelp is a public, high-volume, logistics-heavy support queue. Three things about it shape what a useful agent looks like.

Most inbound is a status question, not a problem to solve. "Where is my parcel", "it says delivered and it isn't here", "I was charged twice". The brand's own reply is usually a short apology plus one instruction or link. An agent that produces exactly that isn't being lazy, it's matching the job.

The brand rarely hides behind DMs, but it does hand off. 10.8% of first replies send the customer somewhere else (0.8% to DMs, 9.9% to a contact page). That's low enough that "draft a grounded reply" is a real task here, unlike Uber_Support at 75%, but high enough that a system which learns to imitate this brand will learn to hand off about one time in ten.

The expensive mistake is public and permanent. A needless escalation costs an agent thirty seconds. A wrong auto-reply is a public tweet from a brand account, and the cases where it goes wrong (fraud, a damaged item, someone who has already been let down twice) are exactly the cases where the customer is least able to absorb it.

So "good" here means:

1. Correct routing matters far more than correct replies. The value of the system is in knowing what it shouldn't answer.
2. Grounded beats fluent. A vague-but-true reply is better than a confident invented one, and the rubric enforces that: fabricated specifics score below honest hedging.
3. Auditable. A support lead has to be able to ask "why did this escalate?" and get a rule name back, not a vibe.
4. Coverage is the business case, harm rate is the constraint. So the headline is an operating point, *X% auto-handled at $\leq Y%$ false-auto*, not an accuracy figure.

What I deliberately didn't build

- **Multi-turn dialogue.** The agent acts on the first customer message only. Later turns are kept for context, and for measuring whether the brand's answer ended the conversation, but conversational state is out of scope.
- **Live tool use.** No order lookup, no account access. This is the biggest cap on the ceiling here: a large share of these tickets simply can't be answered without account data, and no amount of prompt work fixes that. Recognising those and escalating them is the correct behaviour, and it's what the system does.
- **PII redaction.** PII is detected and used as an escalation trigger. It isn't scrubbed or stored.
- **Fine-tuning.** Retrieval plus a taxonomy gets most of the value for a fraction of the complexity, and it stays inspectable.
- **Multilingual support.** Non-English traffic is filtered out of the corpus. It exists in the queue; this system doesn't serve it.
- **Serving.** No API, no queue consumer, no latency budget. The deliverable is an evaluated system, not a deployed one.

2. Method

Stage	Approach
Brand choice	Profiled 7 candidate brands on handoff rate, reply substance, intent diversity, multi-turn rate (results/brand_profile.md)
Corpus	8,496 threads kept from an 11,000-thread sample (957 unusable, 1,530 non-English, 17 near-duplicates dropped), stable-hash split into 7,998 retrieval history / 498 evaluation pool
Taxonomy	Cluster → LLM names each cluster → human merges into taxonomy/intents.yaml, which is both the classifier prompt and the annotator guideline
Golden set	150 adjudicated examples in 3 strata (60% traffic-proportional, 25% rare-intent, 15% hard cases), scored separately, dev 70 / test 80 by stable hash
Agent	retrieve(k=5) → classify → draft (≤280 chars, cites the precedent it used) → route
Routing	Deterministic rules first, LLM only judges what survives them; every escalation carries the rule that caused it
Evaluation	Bootstrap CIs, coverage-vs-harm curve, LLM judge validated against blind human scores
Models	qwen3:4b (local) classifies and drafts, gemini-3.5-flash-lite judges, nomic-embed-text embeds from the committed cache

The brand decision, and a correction

brand	threads	dm defl.	link handoff	handoff	substantive	multi-turn	lexical div.
hulu_support	3976	0.006	0.052	0.057	0.927	0.383	0.124
AmazonHelp	3653	0.008	0.099	0.108	0.716	0.597	0.183
SpotifyCares	3937	0.374	0.003	0.377	0.555	0.355	0.135
XboxSupport	3921	0.287	0.002	0.289	0.547	0.520	0.119
Delta	3950	0.200	0.001	0.201	0.409	0.354	0.157
AppleSupport	3949	0.536	0.035	0.551	0.394	0.350	0.130
Uber_Support	3953	0.648	0.119	0.750	0.170	0.357	0.126

My guess going in was SpotifyCares. The data rejected it at 37.7% handoff.

The first version of this table was wrong. It only measured DM-style deflection and put AmazonHelp at 0.008. Reading actual retrieved replies showed the real pattern: AmazonHelp almost never says "DM us", it says "please reach out to us here: <URL>". Counting both forms moved it to 0.108, a 13× undercount. The ranking survived, so the decision stands, but it stood on a number that was wrong until somebody checked it against raw text. A bare URL deliberately doesn't count as a handoff: roughly half of all replies contain one and most of those are genuinely useful, so the rule requires a contact verb as well.

AmazonHelp over hulu_support is the one call here that is judgement rather than measurement. hulu wins on handoff rate and on substance, but its lexical diversity (0.124) and multi-turn rate (0.383) point at an intent space that collapses to about four classes, which would have made the classification result uninformative.

3. Results

Two parts. The baseline comparison needs the golden labels and is still pending. The reference comparison needs no labels at all, and it is the part that changed what I believe.

3.1 The brand's own reply, scored by the same judge

For all 70 dev rows I put @AmazonHelp's actual reply through the same blind judge, on the same message, against the same retrieved precedent as the agent's draft (eval/reference_replies.py). One model scored all 140 calls (gemini-3.5-flash-lite); providers.switches and degraded_starts are both empty, so this is one judge and not two averaged.

dimension	agent	brand's own reply	delta	comparable?
groundedness	4.39	3.64	+0.74	no — see below
resolution	4.23	3.53	+0.70	yes
tone fit	4.76	3.94	+0.81	yes
safety	4.53	3.90	+0.63	yes
comparable mean	4.50	3.79	+0.72	
would_send	0.87	0.54	+0.33	

Head to head on the comparable dimensions: agent better on 44 rows, human better on 13, tied on 13. Our would_send gate would have blocked **32 of the 70 replies this brand actually posted in public (46%)**.

Read literally, a 4B model on my desktop writes better support replies than the people who answered these tickets. I don't believe that for a second, and working out why is worth more than the table.

Groundedness is excluded from the comparable mean for a structural reason: the human's claims are backed by the order, the account and the tracking page, none of which the judge can see, so the rubric scores them as unsupported. The other three dimensions turned out to be contaminated too, in three ways I could measure.

The judge marks down replies it was handed incomplete. Seven of the 70 historical replies are explicitly part one of two, ending "(1/2)", because the brand splits long answers across tweets. Those score **3.00** on the comparable mean against **3.86** for the rest. The judge is reading half a reply and scoring it as a whole one.

It punishes correct support behaviour it cannot verify. Four replies tell a customer to stop posting tracking details on a public timeline (threads 288010, 2316996, 420919, 1219160). They average **2.58**. On one, the judge wrote that the reply "invents an ungrounded policy not found in any precedents" and "would embarrass the brand". My own route.py hard-escalates any message containing public PII for exactly the reason that human agent gave. The system's safety rule and the system's judge disagree about the same behaviour, and only one of them is right.

It rewards problem-shaped boilerplate. Thread 2750418: the customer tweets a photo of a Prime Now delivery that arrived at their hotel, pleased. The human replied in kind. The agent replied *"I'm sorry you haven't received your Prime delivery to Orlando"*, inventing a complaint that isn't there, and the judge scored the agent **3.33 higher**. Two more of the same shape (2687680, 2619718). That is an agent failure the judge recorded as a win.

So the honest headline of this table is not that the agent is better. It is that an LLM judge with no account access, no thread context and no world knowledge systematically prefers grounded-sounding boilerplate to real support work. Every reply-quality number in this report inherits that bias, which is the strongest argument I have for why the judge-vs-human agreement statistic, not the rubric's apparent rigour, is what licenses any claim about reply quality.

One more thing this measured that the report would otherwise have had to argue. The rubric says a handoff scores 1 on resolution. Six of the 70 historical replies are handoffs, and the judge gave them **4.00** on resolution against 3.48 for direct replies. Small sample, opposite sign: the judge is not applying the rule the rubric states.

A note on two numbers that look inconsistent. The judge would send 87% of the agent's replies, but the router only auto-handles 31 of 70. Different questions: the judge scores the text, while the router also escalates on intent class (18 rows hit never_auto_intent), on thin precedent (13 hit low_retrieval_similarity) and on hard rules (4 safety, 3 payment-card PII, 1 account compromise). The gate is deliberately not a function of reply quality.

3.2 Against the baselines

Pending run. From results/eval_test.md once the golden set is adjudicated and make eval-test has run once. Nothing is estimated ahead of it.

Five systems are compared on the same rows, with bootstrap 95% CIs on every headline number:

System	Intent	Reply	Routing
trivial_always_auto	majority class	one canned string	never escalates
trivial_always_escalate	majority class	—	always escalates
simple_tfidf_nn	TF-IDF + logistic regression	nearest historical reply, verbatim	thresholds only, no LLM
agent_no_retrieval	LLM	LLM, no precedent	full router
agent	LLM + retrieval	LLM, grounded in precedent	full router

The brief asks for two baselines. The third one, the ablation, is the one I care about most: it is the same LLM pipeline with retrieval switched off, so the gap between it and `agent` is what grounding actually buys, separated from what the language model would have done anyway. The trivial and simple baselines answer a different question, which is whether the dataset is easy.

Two reporting choices matter for reading that table. Routing is reported as coverage and false-auto rate rather than accuracy, because a needless escalation and a bad public auto-reply cost wildly different amounts and one accuracy number averages them together. And the learned baselines are scored out-of-fold when the split being scored is the split they were fitted on: in-sample, `simple_tfidf_nn` returns 1.000 intent accuracy by memorisation, which would make the agent look hopeless against a lookup table.

4. Failure analysis

The full top five needs the golden labels, since ranking failure modes by frequency means knowing which rows are wrong. Three are already visible in the reference run, with real thread ids.

1. The agent assumes a complaint when there isn't one. Threads 2750418, 2687680, 2619718. Retrieval returns the five nearest past messages, and on this queue almost every past message is a complaint, so a neutral or positive tweet gets grounded in complaint precedent and comes back as an apology for something that never happened. Hypothesis: a retrieval problem rather than a drafting one. The drafter is doing what it was told with the evidence it was given, and a similarity floor tuned for "is there precedent" does not catch "is this precedent the same *kind* of message".

2. Thin precedent drives a third of all escalations. 13 of 70 rows escalate on `low_retrieval_similarity` and 18 on `never_auto_intent`. The first is the system working as designed. It is also the ceiling on coverage, and no prompt change moves it: only a better index or more history would.

3. The judge's blind spots become the agent's scoreboard. Covered in 3.1: fragments penalised 0.86, PII enforcement scored 2.58, invented complaints rewarded. It belongs here because it is a failure of this evaluation rather than of the brand, and it is the one I would fix first.

Remaining modes pending the labelled set.

5. What is misleading about my headline number?

The brief makes this section mandatory, and it's the one I'd read first.

1. There is one annotator, so the agreement figure is intra-annotator. The κ for label quality comes from re-labelling 50 examples blind, at least a day later. That measures self-consistency, so it's a ceiling on label quality, not inter-annotator agreement. A second annotator would almost certainly agree less. Every metric built on these labels has more slack in it than its confidence interval suggests, because the CI covers sampling noise and not label noise.

2. The golden set was pre-labelled by the system it evaluates. Weak labels came from the classifier and were adjudicated by hand. The override rate is published for exactly this reason: the lower it is, the more the golden set measures agreement-with-the-model rather than ground truth. No amount of care removes that anchoring completely.

3. The language filter leaks, and I know by how much. 10 of 8,496 corpus threads (0.12%) are non-English messages that got through because Latin brand names dominate the token count. One of them is in the golden set. I left it: fixing it means rebuilding the corpus, the clustering, the taxonomy and the weak labels to move 0.12% of the data.

4. $n = 80$ on the test split, so the intervals are wide. The golden set is 150 rather than 250 because the compute budget was a local 4B model. That splits 70 dev / 80 test. Differences under roughly 10 points between systems aren't

distinguishable from noise at this sample size, so any ranking that depends on a small gap should be read as "not established" rather than "smaller".

5. The headline is one stratum, and the pooled number isn't production performance. The golden set deliberately over-samples rare intents (25%) and hard cases (15%). Pooling understates real-traffic accuracy; reporting only the proportional stratum hides the tail. Both are given, but any single number quoted out of this report will mislead in one direction or the other.

6. Historical replies are what the brand did, not what was good. Retrieval grounds the agent in precedent, including precedent where the brand handed off, apologised without acting, or answered a different question. An agent that imitated this queue perfectly would hand off about 10% of the time. The rubric scores deflection as a resolution failure so we don't reward that, which does mean reply scores are measured against a standard the brand's own historical replies don't always meet.

7. The escalation labels are my judgement, not Amazon's policy. "Should a human handle this?" was decided by me against a written guideline. A real support organisation has staffing, SLA and liability constraints that would move that line. The routing metrics measure agreement with my policy, not correctness against a real one.

8. The judge is a different family from the drafter, but it's still a small model. Drafter is Qwen, judge is Gemini falling back to Gemma: different vendors, different weights, so same-model self-preference is mostly removed by construction rather than merely measured. What's left is that a small judge is a weak judge. Its agreement with a human is measured, and that number is what licenses any reply-quality claim here, not the apparent rigour of the rubric.

9. The hard-case slice was designed by the same person who built the agent. It uses model-independent surface heuristics (length, caps ratio, emoji count, rage markers, multi-intent connectives) precisely to limit that, but which heuristics count as hard is still my choice.

10. Thresholds are fitted on ~70 dev examples. `tau_confidence` and `tau_similarity` are point estimates from a small sample. They won't transfer cleanly to a new period or a new brand, and the operating point they define is more fragile than one reported coverage number implies.

11. The drafter is a small cheap model, so absolute quality isn't the architecture's ceiling. Reply quality reflects qwen3:4b running locally, which I chose because ~1000 calls per full run exceeds every free hosted budget available to me, not because it writes the best reply this pipeline could produce. Comparisons between systems are still fair, since every system uses the same generator. Swapping in a stronger drafter would move reply quality, and should, but it wouldn't validate the routing, which is where the value is.

12. The confidence score isn't calibrated. Routing partly depends on a self-reported LLM confidence, which is known to cluster high and behave more like a fluency signal than a probability. It's used because having no confidence signal is worse, not because it's trustworthy.

13. The retrieval backend almost entirely determines what the agent sees. I ran two embedding backends over the same corpus and the same 120 queries. They agree on the single most similar precedent 7.5% of the time, and their top-5 sets overlap 11.3%. So "grounded in how the brand historically resolved this" really means grounded in whatever neighbours the encoder happens to surface, and a different encoder would ground the same reply in almost entirely different evidence. Both backends return plausible precedent when you look at them, which is what makes this easy to miss: spot-checking retrieval and finding it fine does not establish that it is stable. Every reply-quality number here is conditional on this one choice.

14. One brand, one snapshot. All of this is AmazonHelp in late 2017. Nothing here establishes that the taxonomy, the thresholds or the judge's behaviour transfer to another brand or another year.

6. What I'd do with one more week

Ranked by how much each would change what I currently believe:

- 1. A second annotator on 100 examples.** Real inter-annotator κ would replace the weakest claim in this report with a measured one. Highest value by a distance.
- 2. Human preference over pairwise comparisons.** Absolute 1–5 judge scores are the least reliable part of the reply evaluation. Pairwise A/B against the real historical reply is more robust, and it answers "is this better than what the

brand actually sent?" directly.

3. **Calibrate the confidence signal.** Fit a reliability curve on dev, then route on calibrated probability instead of raw self-report. The routing thresholds currently sit on an uncalibrated number.
4. **Transfer test on a second brand.** Re-run the whole pipeline on hulu_support without re-tuning anything. Whatever degrades is the part that was overfitted to AmazonHelp.
5. **Cost and latency accounting.** Three LLM calls per message is a real operating cost and nothing here measures it. A cheaper router, rules only for the easy majority, is probably most of the value.
6. **Retrieval quality ablation.** Vary k, compare embedding backends, and measure how much reply quality actually depends on retrieval depth.